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History of burial, carbonate diagenesis, and exhumation of the Eocene Xianqian Basin in western central Tibet

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ABSTRACT

Lacustrine carbonates in intermontane basins have the potential to archive paleoelevation and paleoclimate information of orogenic belts. Carbonate oxygen and clumped isotopes are commonly used to reconstruct paleo-meteoric water isotopic compositions and depositional temperatures; however, post-depositional processes, including water-rock interaction and solid-state reordering, can introduce uncertainties that complicate such interpretations. To evaluate the fidelity of lacustrine carbonate records, we combine clumped isotope thermometry with apatite thermochronology to constrain the thermal history and carbonate diagenesis in the Xianqian Basin, central Tibet, where >2 km of strata accumulated between 50 and 43 Ma. Low-temperature thermochronometric data indicate that apatite U-Th/He ages were fully reset, whereas apatite fission-tracks were only partially annealed, suggesting burial temperatures between 60 and 120 °C. These burial temperatures are insufficient to raise plausible near-surface carbonate clumped isotope temperatures to the observed 47–77 °C through solid-state bond reordering alone. Together with petrographic, mineralogical, and geochemical evidence, we interpret the elevated clumped isotope temperatures as the result of dolomitization during the shallow-burial stage. Consequently, the calculated oxygen isotope compositions reflect those of diagenetic fluids ($\delta^{18}O_w$ = +1‰ to +6‰), but not those of surface freshwater required for paleoelevation reconstruction. This study underscores (1) the value of integrating low-temperature thermochronology with clumped isotope thermometry to resolve carbonate diagenetic histories; and (2) the need to re-evaluate previous conclusions that the Qiangtang terrane had reached near-present elevations during the Paleogene.

1. Introduction

Constraining the paleoelevation history of orogenic belts is critical for understanding how deep-Earth processes drive surface uplift and how mountain building influences climate, drainage systems, and ecosystems (Poage and Chamberlain, 2001; Mulch, 2016; Dupont-Nivet et al., 2025; Spicer et al., 2025). Carbonates formed in continental settings, such as paleosol nodules and lacustrine micrites, are valuable archives of the isotopic composition of ancient surface waters (Quade et al., 2020). The fractionation factor of oxygen isotopes between carbonate mineral and water is controlled by temperature (Kim and O'Neil, 1997; Horita, 2014), which can be independently constrained by clumped isotope thermometry. When clumped isotope-derived temperatures reflect near-surface conditions, both the formation temperature and the calculated water $\delta^{18}O$ ($\delta^{18}O_w$) can be used to estimate paleoelevation, assuming reasonable paleo-temperature lapse rates (e.g., Huntington et al., 2010; Li et al., 2024b) and the elevation gradient of stable isotopes in precipitation, respectively (Poage and Chamberlain, 2001; Rowley, 2007; Krsnik et al., 2021).

However, carbonate minerals are susceptible to post-depositional

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alteration, which can elevate clumped isotope temperatures above Earth's surface temperature, change the original $\delta^{18}O$ values, and also alter the elemental and mineralogical signatures (Choquette and Pray, 1970; Immenhauser, 2022; Li et al., 2024a, 2024c). For carbonates with clumped isotope temperatures higher than surface temperature, the thermal history of the basin, such as that obtained from low-temperature thermochronology, can be used independently to model solid-state bond reordering (Hemingway and Henkes, 2021) and evaluate whether the elevated temperatures could have resulted from water-rock interaction. Petrography, mineralogy, and elemental geochemistry are also important for evaluating the source of diagenetic fluids (e.g., Smedley and Kinniburgh, 2023; Li et al., 2024a, 2024c).

Over the past several decades, the paleoelevation history of the Tibetan Plateau has been a research focus due to its very high mean elevation (~5 km) and first-order orographic effect on global climate (Molnar et al., 1993; Ding et al., 2022). Recent studies suggested that during the Eocene, south-central Tibet was a "proto-plateau" consisting of local topographic highs, i.e., Lhasa and Qiangtang terranes, separated by an intermontane valley in central Tibet of 0–3 km paleoelevation along the Bangong-Nujiang suture zone (Wei et al., 2016; Ding et al., 2022; Li et al., 2022; Xiong et al., 2022; Fig. 1A). Although recent reports of fish, insect, and plant fossils support the existence of such a low-elevation valley (Wu et al., 2017; Cai et al., 2019; Su et al., 2020), it remains debated how pristine the oxygen isotopes of carbonates are in these basins. In the Gerze Basin, the $\delta^{18}O$ values of carbonates (−8‰ to −2‰, VPDB) were interpreted as primary signals related to marine excursion (Wei et al., 2016), but a recent investigation of the clumped isotopes suggested that they recorded high temperature (40–50 °C) resulting from dolomitization and thus the $\delta^{18}O$ may have been altered (Li et al., 2024a). In the Nima Basin, the late Oligocene paleosols were interpreted to have undergone either cryptic recrystallization in a rock-buffered system (Huntington et al., 2015) or solid-state reordering (Li et al., 2024c), leaving the reliability of the oxygen isotopes of paleosols undetermined (DeCelles et al., 2007). The thermal histories of these basins were argued to be important for the interpretation of the clumped isotope data, but have only been loosely constrained by estimates of stratal thickness, structural relationships, and geothermal gradient (Huntington et al., 2015; Li et al., 2024a, 2024c).

North of the Lhasa-Qiangtang suture valley, thermochronologic and geomorphic evidence suggests that the Qiangtang terrane may have reached plateau-like conditions and/or near-modern elevations by ~45–40 Ma (Rohrmann et al., 2012; Law and Allen, 2020), but quantitative paleoelevation constraints remain sparse (Xu et al., 2013; Shen et al., 2023). Compared to central and southern Tibet, the diagenetic evolution of lacustrine carbonate in the Qiangtang terrane remains poorly understood, raising great uncertainty for reconstructing its surface uplift history.

The Xianqian Basin in southwestern Qiangtang (Fig. 1A) preserves one of the best-exposed Eocene redbed successions in central Tibet, including abundant interbedded tuffs and carbonates (Ma et al., 2026). These strata provide an excellent opportunity to investigate carbonate diagenesis and evaluate its implications for paleoelevation

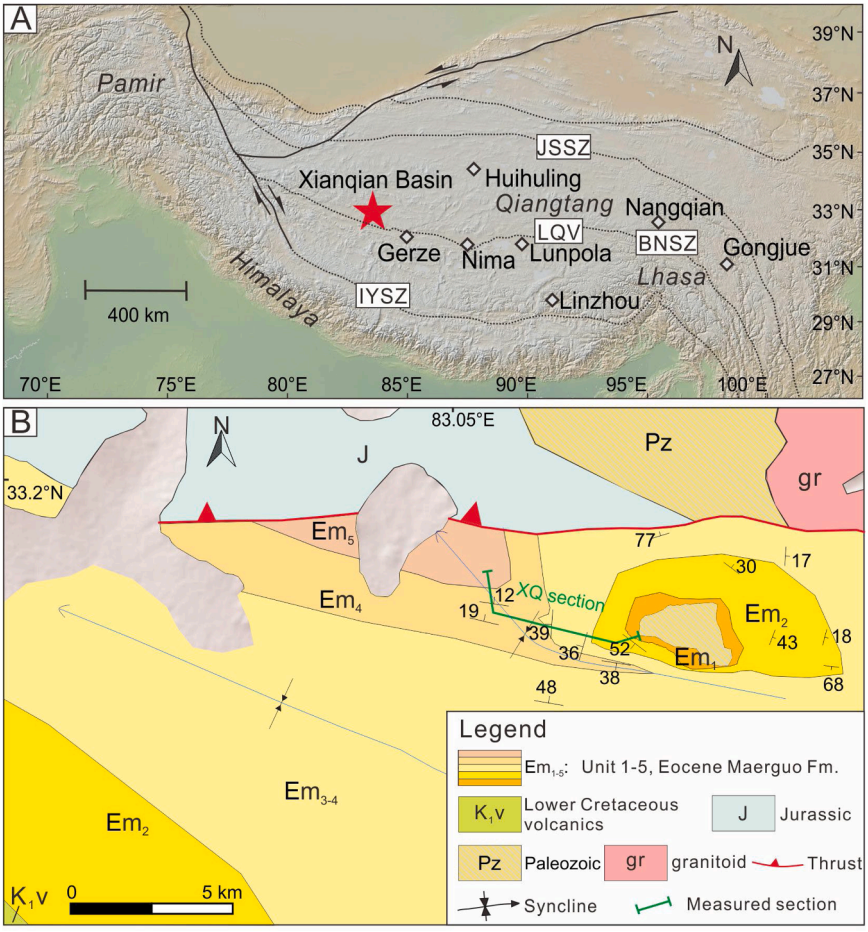


Fig. 1. (A) Location of Xianqian Basin in Tibetan Plateau, with base topographic map from GeoMapApp. Other Paleogene basins, including the Huihuling, Nangqian, and Gongjue basins in central Qiangtang, the Gerze, Nima, and Lunpola basins along the Bangong-Nujiang suture zone, and the Linzhou Basin in Gangdese arc in southern Tibet, are plotted for reference. IYSZ: Indus-Yarlung suture zone; BNSZ: Bangong-Nujiang suture zone; LQV: Lhasa-Qiangtang suture valley; JSSZ: Jinsha suture zone. (B) Geological map of the northern Xianqian Basin, showing the distribution of stratigraphic units and the location of the measured section.

reconstruction of the Qiangtang terrane. In this study, we characterize the diagenetic history of Xianqian Basin lacustrine carbonates and assess their suitability for stable and clumped isotope-based paleoaltimetry.

2. Geological background

The Qiangtang terrane in the central Tibetan Plateau is bounded by the Jinsha suture to its north and Bangong-Nujiang suture to its south (Fig. 1A). It consists of high-pressure metamorphic rocks in its core and is flanked by mostly Paleozoic to Mesozoic shallow marine strata in the north and south (Kapp et al., 2005; Ma et al., 2023a). The Mesozoic basins experienced sea disappearance during Late Jurassic to Early Cretaceous time and were uplifted during mid-Cretaceous time due to collision between the Lhasa and Qiangtang terranes (Ma et al., 2023a, 2023b). During the Cenozoic, localized intermontane basins developed across the Qiangtang terrane (Wu et al., 2008; Xu et al., 2013; Ma et al., 2026), preserving important sedimentary archives for paleoelevation reconstructions.

One such archive is the Xianqian Basin, which contains ~2000-m-thick redbeds of the Maerguo Formation (Ma et al., 2026). This basin was situated ~100 km to the north of the Lhasa-Qiangtang suture valley (Fig. 1B). Zircon U-Pb geochronology of interbedded tuffs indicates that sedimentation within the basin was short-lived from ~50–43 Ma (Fig. 2, Ma et al., 2026). Five stratigraphic units were recognized in the Maerguo Formation. In the measured Xianqian section, Unit Em_1 (Lower Conglomerates, 49.6–49.1 Ma) unconformably overlies Mesozoic granitoids and mainly consists of alluvial fan-dominated conglomerate and minor sandstone with a total thickness of 83 m. Up section, Unit Em_2 (Fluvial Redbeds, 49.1–47.0 Ma) consists of 385 m of sandstone and mudrock deposited in fluvial environments. Unit Em_3 (Lower Fine-grained Lacustrine Redbeds, 47.0–44.7 Ma) consists of 461 m of mudrock, sandstone, and carbonate deposited in shallow lacustrine environments. Unit Em_4 (Upper Lacustrine Unit, 44.7–42.8 Ma) consists of 794 m of grey sandstone, mudrock, and carbonate deposited in delta front to prodelta. Finally, Unit Em_5 (Upper Conglomerates, 42.8–42.6 Ma) consists of >136 m conglomerate and minor sandstone deposited in alluvial fan environments (Fig. 2).

3. Samples and methods

To reconstruct the thermal history of the Maerguo Formation, we conducted apatite (U-Th)/He (AHe) dating of one tuff sample from the bottom of the formation and detrital apatite fission track (AFT) analysis of four sandstone samples from different stratigraphic levels. Sixteen carbonate samples were collected from the two lacustrine units (Em_3 and Em_4) of the Maerguo Formation (Fig. 2). Petrographic microscopy, cathodoluminescence analysis (CL), electron probe microanalysis (EPMA) elemental mapping, X-ray diffraction (XRD), and inductively coupled plasma mass spectrometry (ICP-MS) were employed to characterize the petrography, mineralogical composition, and element

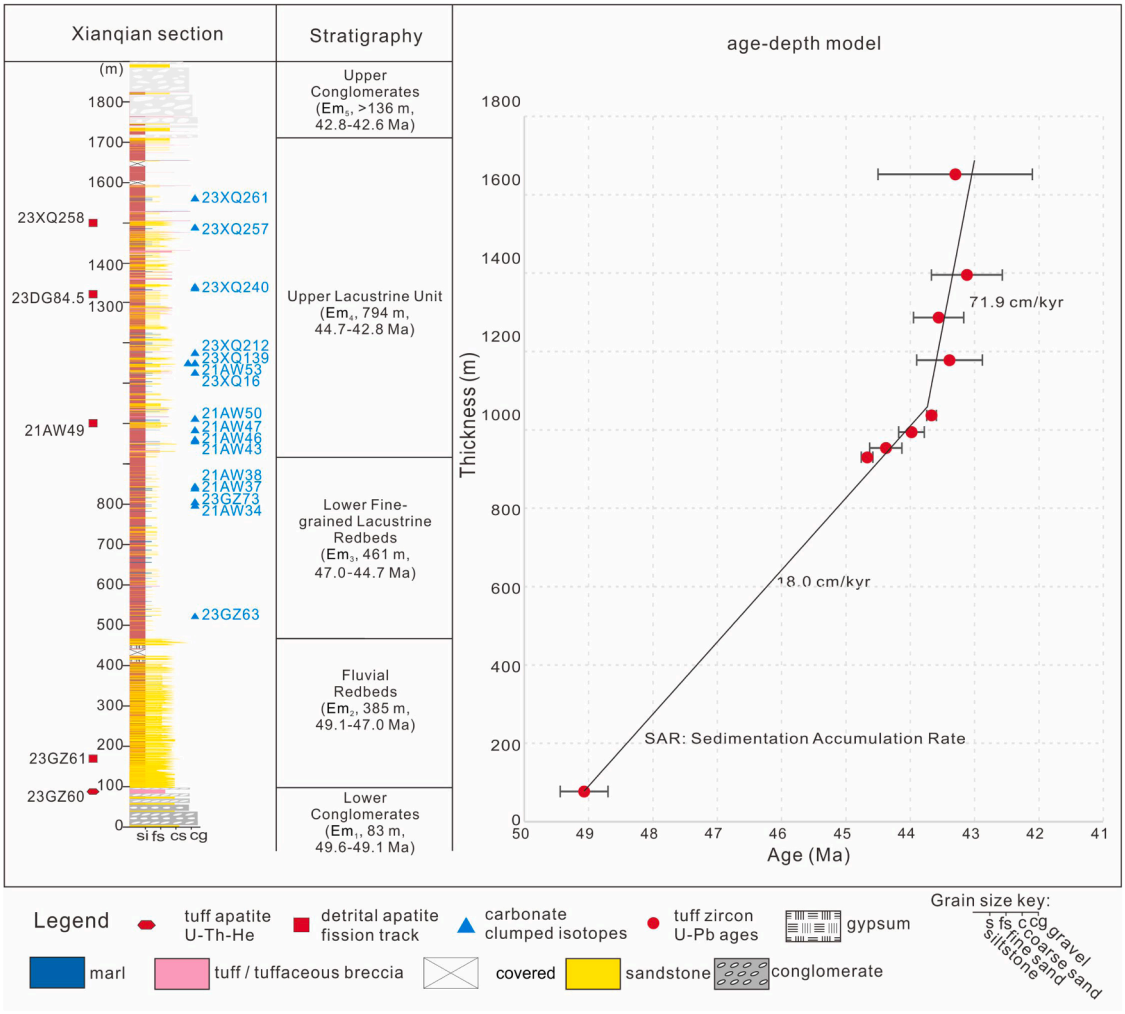


Fig. 2. Measured log of the Xianqian section showing the five units of the Maerguo Formation, with carbonate, tuff, and sandstone samples indicated alongside the log (Ma et al., 2026). The age-depth model of the Maerguo Formation on the right is adapted from Ma et al., 2026.

composition and distributions of these lacustrine carbonates. Clumped isotopes and carbon and oxygen isotopes were measured to reveal the temperature of carbonate (re)crystallization and the isotopic compositions of fluids from which the carbonate grew. To further evaluate the potential diagenesis, modeling of solid-state bond reordering was conducted using the independently constrained thermal history of the basin.

3.1. Apatite (U-Th)/He (AHe) dating

Apatite grains were concentrated using standard heavy liquid and magnetic separation procedures. For U–Th/He analysis, five subhedral to euhedral inclusion-free apatite grains were hand-picked after examination under a binocular microscope. Grain dimensions were measured prior to analyses, and photographs were taken of each grain. Apatite aliquots were loaded into platinum capsules and outgassed under vacuum at ~900 °C for 5 min using a fiber-optically coupled diode laser with an 820 nm wavelength. ^{4}He abundances were determined as an isotope ratio using a pure ^{3}He spike calibrated against an independent ^{4}He standard, with the uncertainty in the ^{4}He measurement estimated at <1%. Apatite U–Th–Sm dating was performed on an Agilent 7700 quadrupole ICP-MS, and the same spike amounts were treated as unspiked reagent blanks. A second digestion in HCl for 24 h at 200 °C ensured the dissolution of fluoride salts, and the final solutions were diluted to 10% acidity for analysis on the Agilent 7700 quadrupole ICP-MS. Analytical uncertainties for the U–Th/He facility at State Key Laboratory of Continental Dynamics (Northwest University) are estimated at ~6.2% (±1σ), which incorporates the uncertainty associated with the alpha-correction, an estimated 5 μm uncertainty in grain size measurements, as well as uncertainties from gas analysis and the ICP-MS. Durango apatite and Fish Canyon Tuff zircon were measured as internal standards and used as additional monitors for analytical precision. Ages were corrected for alpha ejection (F_T correction; Farley et al., 1996). The AHe data are presented in supplementary Table S2.

3.2. Apatite fission track (AFT) analysis

Apatite grains were mounted in epoxy resin on glass slides, ground, and polished to expose their internal surfaces at the State Key Laboratory of Critical Earth Material Cycling and Mineral Deposits, Nanjing University, China. The polished mounts were etched in 5 M HNO_3 for 20 s at 20 °C to reveal fossil tracks. Apatite grains exhibiting polished surfaces parallel to prismatic crystal faces and homogeneous track distributions were selected using a Zeiss Axio Imager M2m microscope. A series of high-resolution digital images were then captured in both reflected and transmitted light at a total magnification of 1000× using a high-sensitivity, fast camera. Track counts were manually performed using the FastTracks software (Gleadow et al., 2009). Confined track lengths were measured from the captured digital images in c-axis-parallel grains.

The ^{238}U content was measured by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS), using an Agilent 8900 ICP-QQQ coupled to an ESL (Elemental Scientific Lasers) New Wave NWR 193UC (Two-Vol2) laser ablation system at Beijing Quick-Thermo Science and Technology Co., Ltd. Ablation was carried out over 25 s using a 30 μm diameter beam, ~3 J/cm^2 energy, and a 5 Hz repetition rate on the selected grains and reference materials (primary NIST SRM 612 glass and secondary Durango apatite) after approximately 10 s of baseline signal collection. Uranium concentrations were calculated using calcium (assumed to be stoichiometric in apatite) as an internal elemental standard, employing the Iolite data reduction package within the Wavemetrics Igor Pro data analysis software (Paton et al., 2011). Single-grain AFT ages were calculated using an absolute calibration based on primary constants (aggregate $\xi = 2.001 \times 10^{-3}$). The AFT age data are presented in supplementary Tables S3 and S4.

3.3. Thermal history modeling

To reconstruct the thermal history of the Maerguo Formation, we carried out inverse thermal history modeling using the QTQt software (Gallagher, 2012) by integrating AHe and AFT data (ages and confined track lengths). Inverse modeling in QTQt uses a Bayesian Markov Chain Monte Carlo approach where a user can input the thermochronometric data and define t-T constraints. The model outputs are a population of acceptable thermal histories based on a posterior probability (Gallagher, 2012). "Expected" paths represent the weighted mean thermal history from the posterior distribution. AHe data were modeled using the helium diffusion model of Flowers et al. (2009).

3.4. XRD mineralogical analysis of carbonate

The samples were processed into finely ground powders (~200 mesh) using an agate mortar, thus generating unoriented powdered specimens. Then the XRD analyses were conducted using a Rigaku D/MAX RAPID II diffractometer (Rigaku, To-kyo, Japan) equipped with a rotating Mo anode, a 0.3 mm collimator, and a curved imaging plate detector. The instrument was operated under conditions of 50 kV and 90 mA, with a counting time of 10 s. The samples, placed in 0.5 mm Kapton tubes, underwent scanning within a range of 2 to 70° or 2 to 90° 2θ. Subsequent data reduction and interpretation were performed using Jade 6.5 software. Semiquantitative weight percentages (wt %) of each mineral present were estimated using the reference intensity ratio (RIR) available in the XRD database. The wt % is proportional to the (XRD intensity)/RIR. All wt % values were normalized in such a way that the sum of all entries equals 100%. The dolomite ordering was quantified using the intensity ratio of the [015] to [110] reflections [I(015)/I(110)] (Goldsmith and Graf, 1958), and the stoichiometry was determined from the d(104) following the empirical calibration of Lumsden (1979). XRD mineral data are presented in Table S1.

3.5. Cathodoluminescence imaging and EPMA elemental mapping of carbonate

The carbonate samples were cut into thin sections and then imaged in a vacuum setting by a cathodoluminescence (CL) Cold Cathode CL8200MK5 in Nanjing University, with a working voltage of 10.5 kV and a current of 250 μA. Three carbonate samples, from the bottom of Unit Em_3 (23GZ73), the top of Unit Em3 (21AW34) and the top of Unit Em_4 (23XQ257) of the Maerguo Formation, were selected for elemental mapping using EPMA. Thin sections were finely polished, carbon coated, and then analyzed at the Kuiper Imaging Facility at State Key Laboratory of Nuclear Resources and Environment, East China University of Technology, China, using a Cameca SX- 100 Ultra instrument.

3.6. Major and trace element analyses of carbonate

> 50 mg of powdered carbonate-rock samples were carefully microdrilled, weighed, and dissolved overnight in 3 mL of distilled 0.5 M acetic acid. Nitric and hydrochloric acids were not used to avoid contamination by weathering detrital materials. The solutions were centrifuged, and the supernatant was transferred to another beaker. The residues were rinsed three times using ultrapure water, and the solutions were added to the previous supernatant. The leachates were converted to nitric acid medium, which was used for elemental analysis carried out at Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences. Major element concentrations were measured using a Thermo Fisher Scientific iCAP 6500 ICP-OES. Trace-element concentrations were determined using an Agilent ICP–MS with analytical precision of ±2% based on repeated analysis of the international reference standard JDo-1 (Imai et al., 1996). The carbonate element concentration data are presented in Table S1 and supplementary Table S5.

3.7. Carbonate clumped and carbon-oxygen isotopes

About 2 mg of powdered carbonate was reacted with dehydrated phosphoric acid under vacuum at 70 °C. The conventional isotope ratio measurements (relative to VPDB) were calibrated using repeated measurements of International Atomic Energy Agency (IAEA) reference materials. The clumped isotope composition of the carbonates (i.e., ^{16}O-^{13}C-^{18}O bonds, denoted as Δ_{638}) in excess of the stochastic distribution was measured at the Environmental Isotope Laboratory at the University of Arizona using a tunable infrared laser differential spectrometer (TILDAS). The term Δ_{638} is the laser measurement equivalent of Δ_{47} in mass spectrometry-based analyses, with a precision of ±0.021‰. The carbonate $\delta^{18}O_c$ and $\delta^{13}C_c$ values were also determined during the measurement, with precisions of ±0.04‰ and ±0.03‰, respectively.

The Δ_{638}–temperature relationship for carbonates at relatively low temperatures is based on 44 synthetic calcite samples equilibrated at temperatures ranging from 6 to 70 °C (Yanay et al., 2022):

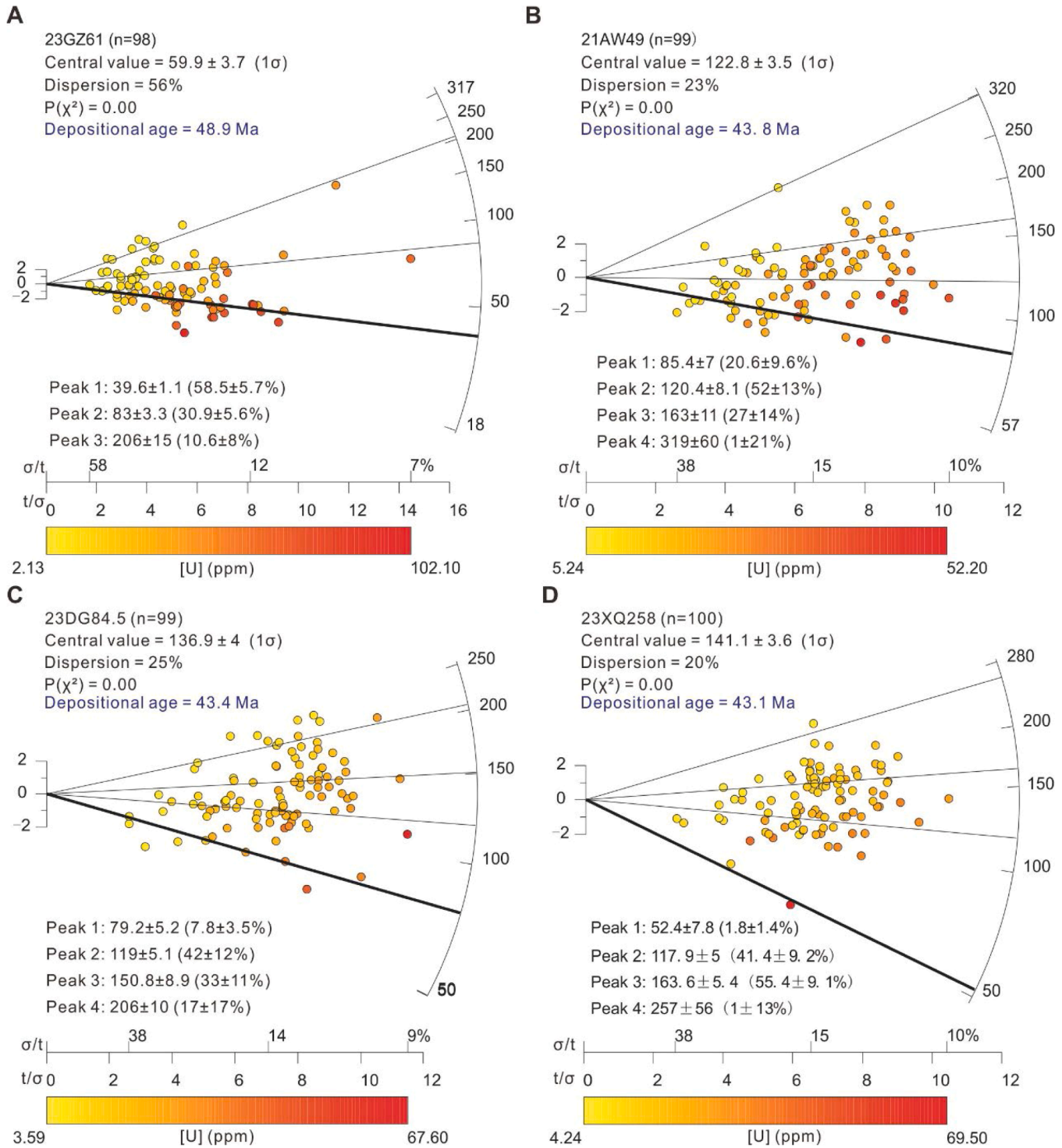


Fig. 3. Radial plots of detrital AFT ages for samples 23GZ61 (A), 21AW49 (B), 23DG84.5 (C), and 23XQ258 (D), constructed using RadialPlotter (Vermeesch, 2012).

$$\Delta_{638CDES} = (0.0405 \pm 0.0006) \times \left(10^6 / T^2\right) + (0.1822 \pm 0.0061), R^2 = 0.985$$

The $\delta^{18}O_w$ values of the water in which the carbonates precipitated were calculated from the clumped isotope temperatures and $\delta^{18}O_c$ values using a linear combination of the dolomite calibration of Horita (2014): $10^3 ln\alpha_{(dolomite\text{-}water)} = (3.14 \times 10^6)/(T + 273.15)^2 - 3.14$; and calcite calibration of Kim and O'Neil (1997): $10^3 ln\alpha_{(calcite\text{-}water)} = (18.03 \times 10^3)/(T + 273.15) - 32.42$. Detailed method description can be found in Li et al. (2024a). Supplementary Data S1 includes the clumped isotope raw data of all samples.

3.8. Solid-state bond reordering modeling

Potential carbonate clumped isotope bond reordering due to burial (heating) was evaluated for samples from the Unit Em_3 and Unit Em_4 of the Maerguo Formation using the Isotopylog package for Python of Hemingway and Henkes (2021). We used the kinetic parameters of dolomite and calcite in Hemingway and Henkes (2021). The thermal history (produced T-t data with QTQT software) for the Maerguo Formation, inverse modeled based on the AFT-AHe (samples 23GZ60 from the Unit Em1 and 23GZ61 from the Unit Em_2) data, was input ifor the carbonate solid-state reordering modeling.

4. Results

4.1. AHe and AFT thermochronology

Tuff sample 23GZ60 (zircon U-Pb age of ~49 Ma; Ma et al., 2026) from the top of Unit Em_1 (at ~89 m in the measured section) of the Maerguo Formation was analyzed for AHe dating. Of the five grains analyzed, four yielded reproducible AHe ages of 36.5 ± 2.4, 39.9 ± 2.8 Ma, 40.4 ± 2.8 Ma, and 39.6 ± 2.8 Ma (1σ), with effective uranium (eU) concentrations ranging from 5.2 to 10.3 ppm (Table S2). One grain yielded younger AHe ages of 22.4 ± 1.5 Ma, with higher eU concentrations of 40.1 ppm.

Four sandstone samples were analyzed for AFT ages: one near the bottom of Unit Em_2 (23GZ61 at 171 m) and three from Unit Em_4 (21AW49 at 1002 m, 23DG84.5 at 1326 m, 23XQ258 at 1496 m). Each sample yielded 98–100 single-grain AFT dates (Table S3). All samples fail the $\chi 2$ test with P ($\chi 2$) =0 (Fig. 3), indicating mixed grain age distributions. 70 lengths of confined tracks were measured for sample 23GZ61, with an average non-projected length of 12.68±1.76 μm (1 sd) and c-axis projected mean track length of 13.58±1.38 μm (1 sd) (Table S4). AFT analysis was not performed on the tuff sample (23GZ60) due to an insufficient yield of apatite grains.

AFT age distributions were deconvolved using RadialPlotter (Vermeesch, 2012). Sample 23GZ61 from Unit Em_2 yields three populations (Fig. 3A) of 39.6 ± 1.1 Ma (P1; 58.5 ± 5.7%), 83 ± 3.3 Ma (P2; 30.9 ± 5.6%), and 206 ± 15 Ma (P3; 10.6 ± 8%), respectively. The P1 age is younger than the depositional age of 23GZ61 (~49 Ma), which was inferred by age-depth modeling (Ma et al., 2026; Fig. 2).

All three Unit Em_4 samples yield four populations, which are all older than their depositional ages between ~44–43 Ma (Fig. 2). The youngest age modes (P1) for samples 21AW49 and 23DG84.5 are 79 Ma (7.8 ± 3.5%) and 85 Ma (20.6 ± 9.6%), respectively (Fig. 3B, C). In contrast, only two single-grain dates in sample 23XQ258 define a P1 age at 52 Ma (Fig. 3D). A consistent P2 population at 117–120 Ma is observed across all three Unit Em_4 samples, while P3 populations range from 150 to 164 Ma. Additionally, sample 23DG84.5 yields a well-defined P4 population at 206 Ma (17 ± 17%). Although few P4 ages of 257 Ma (23XQ258; 1 ± 13%) and 319 Ma (21AW49; 1 ± 21%) are recorded, they are not statistically significant due to the small number of grains (Fig. 3).

4.2. Carbonate petrology, minerology, and element concentrations

Petrographic observations of the studied carbonate samples from the Unit Em_3 and Unit Em_4 show micritic textures (Fig. 4). The massive and laminated micritic dolomite show bright to dull luminescence (Fig. 4D, E). XRD analysis shows that the dolomite percentage in carbonate fractions ranges from 68 to 100% (Fig. 5A). The mol% $MgCO_3$ of dolomites range from 42.4 to 49.2 and the cation orderings range from 0.26 to 0.47 (Table S1). EPMA elemental mapping of Ca, Mg, and Si was conducted on three samples from the bottom, middle, and top of the Xianqian section (Fig. 4A, C). Samples 23GZ73 (near the bottom of Unit Em_3) and Sample 21AW34 (from the upper Unit Em_3) display massive textures, while sample 23XQ257 (near the top of Unit Em_4) shows well-developed lamination.

Uranium (U) concentrations of acetic acid etching materials from ten micritic dolostone samples range from 3.2 to 30.2 ppm. Thorium (Th) contents in six samples range between 1.3 and 8.1 ppm, while the remaining four samples exhibit lower Th values (0.4–0.7 ppm). Mn/Sr ratios (in mg/kg) are consistently elevated, ranging from 1.6 to 31.1 with an average of 15.7. For REE+Y pattern of carbonates, the MREE is slightly enriched compared to LREE and HREE and there exhibits a slightly positive Eu anomaly from 1.1 to 1.2, averaging at 1.1 (Table S1; Fig. S1).

4.3. Carbon, oxygen, and clumped isotopes

Stable isotope analyses of sixteen dolostone samples from the Unit Em_3 and Unit Em_4 yield carbonates $\delta^{18}O_c$ values (VPDB) ranging between − 4.9‰ and − 0.56‰ (Fig. 5B). The $\delta^{13}C_c$ values (VPDB) are more tightly clustered between − 5.7‰ and − 1.7‰, except for one outlier of +2.2‰ (Fig. 5C).

Clumped isotope temperatures ($T_{(\Delta 638)}$) of the sixteen micrite dolostone samples, calculated from Δ_{638} values based on equations in Yanay et al. (2022), range between 47 ± 4 °C and 77 ± 5 °C, which are significantly higher than the Earth surface temperaturess (Table 1). Although a general decrease in $T_{(\Delta 638)}$ is observed upward through the stratigraphic section, the trend is weak with a low R^2 value of 0.2 (Fig. 5D).

5. Discussion

5.1. Thermal history of the Xianqian Basin

If a detrital sample is buried shallower than the partial annealing zone (PAZ, corresponding to a temperature range of about 60–120 °C), its AFT age typically reflects a mixture of the cooling ages of detritus from its source region. In such cases, the cooling age should be older than the depositional age. However, our sandstone sample 23GZ61, collected from Unit Em_2, exhibits a youngest AFT age population (P1) that is younger than its depositional age (~49 Ma). This indicates that the apatite grains experienced at least partial annealing during burial. The presence of multiple age components, however, suggests that complete thermal resetting was not achieved. We thus interpret the sample 23GZ61 to have been buried within the PAZ at 60–120 °C. Because most sources of data dispersion in AHe dating generally skew ages to be older (Fitzgerald et al., 2006; He et al., 2021; Flowers et al., 2023), and the consistency of measured ages in the sample suggests it cooled relatively rapidly, a likely explanation for the one anomalously young date (22.4 Ma) is a high-eU rim resulting in undercorrection of helium lost due to alpha ejection (Ault and Flowers, 2012).

To constrain the thermal history of the basin, we integrated the AHe data of the bottom tuff sample 23GZ60 with the AFT data of its neighboring sandstone sample 23GZ61 (located 82 m stratigraphically apart) for inverse thermal modeling. Thermal history modeling using detrital AFT data, although more challenging than modeling data from a bedrock sample, can still provide useful first-order constraints on the

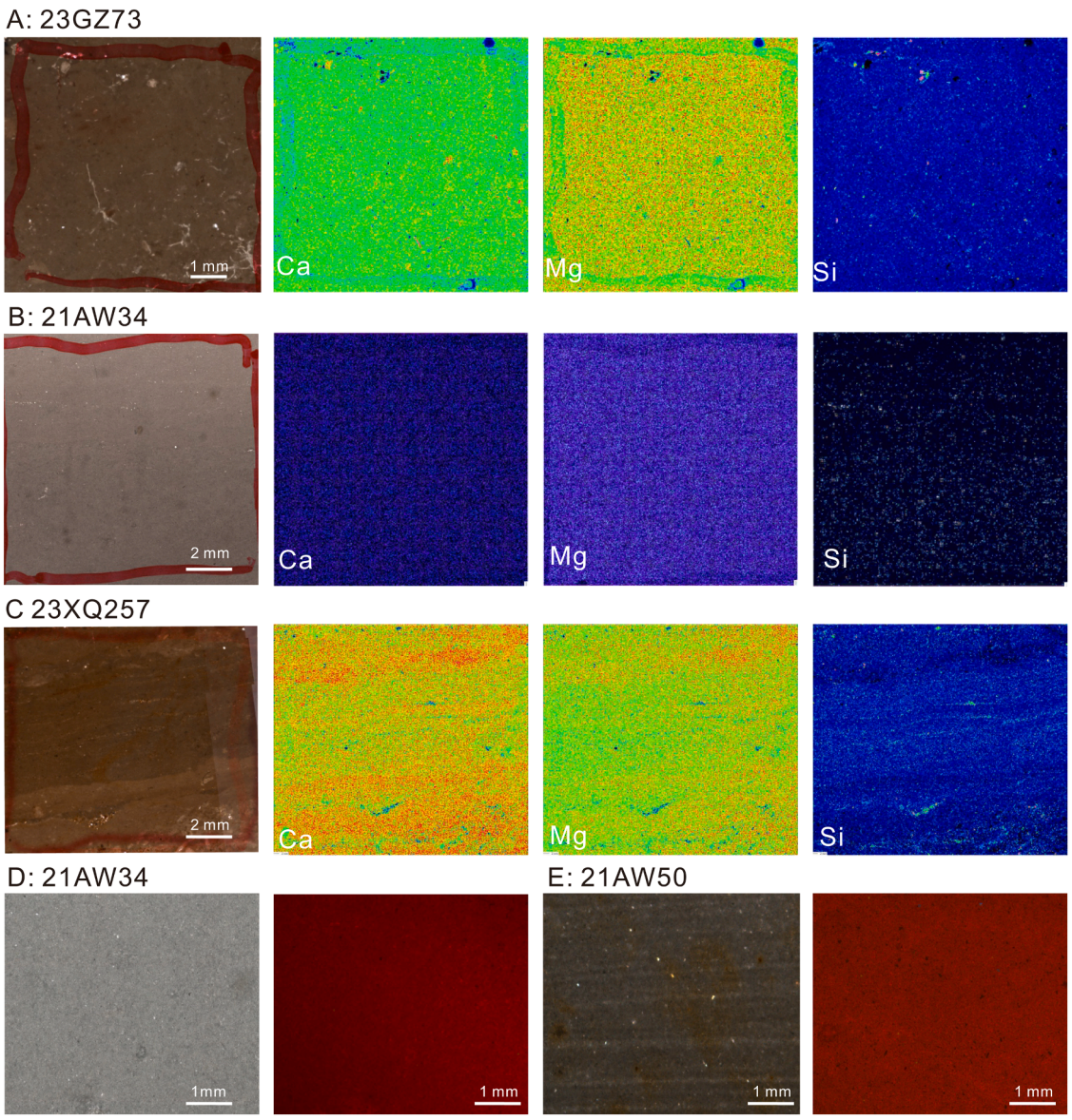


Fig. 4. (A, B, C) Microphotographs and EPMA elemental (Ca, Mg, Si) maps of representative lacustrine carbonate samples 23GZ73 (Unit Em_3,), 21AW34 (Unit Em_3), and 23XQ257 (Unit Em_4), respectively. (D, E) Microphotographs and CL images of representative massive (21AW34, Unit Em_3) and laminated (21AW50, Unit Em_4) carbonate, respectively.

burial and cooling history of a basin when interpreted in combination with independent geological context (e.g., Li et al., 2017). Constraints include a surface temperature of 25 ± 10 °C at ~49 Ma and a modern surface temperature of 10 ± 10 °C. The input data, constraints, and set-up parameters for thermal history modeling are reported in Table S6 based on framework established by Flowers et al. (2015). Our inverse modeling result shows that the basin experienced peak burial temperatures of ~106 °C around ~42 Ma, followed by exhumation to near-surface conditions by ~34 Ma (Fig. 6).

In our models, the predicted AFT age is younger than the observed central age. This discrepancy arises because partial annealing of detrital AFT in the PAZ produces a central age that represents a weighted average of fully and partially reset grains, rather than the actual timing of partial annealing.

The three AFT samples from Unit Em_4 yield ages older than their corresponding depositional ages, implying that these samples were buried to shallower depths than sample 23GZ61 in Unit Em_2. Although the age distributions alone do not allow us to definitively assess whether the AFT systems in these samples experienced partial annealing, we consider partial annealing unlikely. These samples are located 831–1325 m stratigraphically above 23GZ61, and the overlying sediment thickness (400–894 m) would be insufficient to reach the partial annealing zone (PAZ) for AFT (60–120 °C), assuming a geothermal gradient of 30 °C/km and a surface temperature of 25 °C. We therefore interpret the observed AFT age populations, including peaks at ~160 Ma, ~120 Ma, and ~80 Ma, as recording major regional tectonothermal events in central Tibet (Kapp et al., 2005; Ma et al., 2023a; Li et al., 2025).

5.2. *Dolomitization caused by sedimentary burial*

Several mechanisms may account for clumped isotope-derived temperatures that exceed Earth's surface conditions, including solid-state bond reordering, kinetic isotope effects, and water-rock interaction (including recrystallization and dolomitization) (Li et al., 2024b). Solid-state bond reordering due to elevated burial temperature is unlikely because low-temperature thermochronology data indicate the maximum burial temperature of no more than 110 °C (Fig. 6), which is

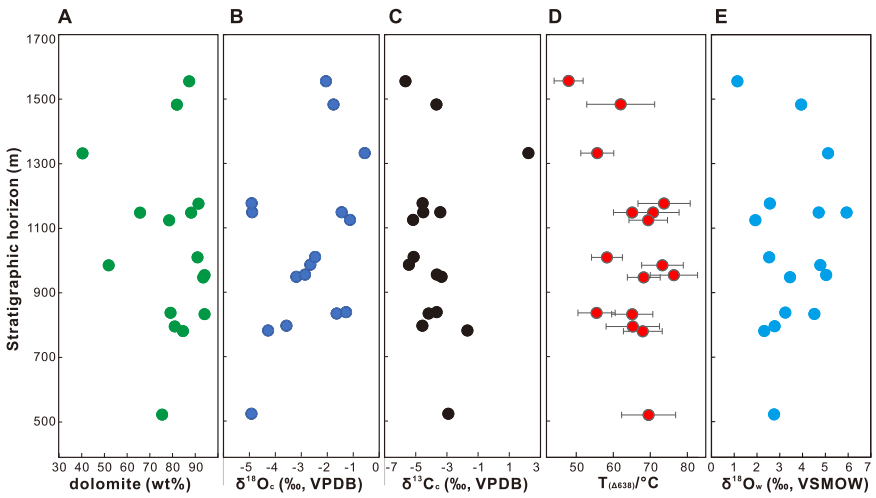


Fig. 5. Stratigraphic variation of dolomite content (A), $\delta^{18}O_c$ (B), $\delta^{13}C_c$ (C), $T_{(\Delta 638)}$ (D) and calculated diagenetic fluid $\delta^{18}O_w$ (E) for lacustrine carbonate samples from the Maerguo Formation. $T_{(\Delta 638)}$ exhibits a general decreasing trend up-section, whereas dolomite content, $\delta^{13}C_c$ $\delta^{18}O_c$, and $\delta^{18}O_w$ show no clear correlation with stratigraphic position.

Table 1
Summary of stable isotope data of lacustrine carbonate in the Xianqian Basin.

sample ID	Rep. no	$\delta^{13}C_c$ (VPDB/‰)	SD (‰)	$\delta^{18}O_c$(VPDB/‰)	SD (‰)	$\Delta 638_{CDES}$ (‰)	SD (‰)	1SE (‰)	95% conf.	Temp.(°C)	error (°C)	$\delta^{18}O_w$ (VSMOW/‰)	Error (‰)
21AW34	5	−1.69	0.02	−4.28	0.01	0.529	0.016	0.002	0.004	68.3	4.3	5.79	0.81
21AW37	5	−4.18	0.00	−1.64	0.02	0.535	0.019	0.002	0.004	65.6	4.2	8.03	0.68
21AW38	5	−3.66	0.02	−1.27	0.02	0.556	0.018	0.002	0.004	55.9	3.9	6.82	0.81
21AW43	5	−3.34	0.02	−3.19	0.05	0.529	0.009	0.002	0.004	68.7	4.3	6.94	0.68
21AW46	5	−3.65	0.01	−2.85	0.01	0.513	0.021	0.002	0.004	76.9	4.6	8.53	0.69
21AW50	5	−5.14	0.01	−2.47	0.03	0.518	0.017	0.002	0.004	58.8	4.0	6.09	0.68
21AW47	5	−5.44	0.01	−2.65	0.02	0.550	0.010	0.002	0.004	73.8	4.4	8.28	0.67
21AW53	5	−4.52	0.01	−1.45	0.02	0.535	0.016	0.002	0.004	65.7	4.2	8.23	0.67
23GZ63	3	−2.91	0.02	−4.92	0.02	0.524	0.021	0.004	0.007	70.9	5.2	5.53	0.69
23GZ73	3	−4.58	0.03	−3.57	0.02	0.535	0.022	0.003	0.007	65.4	5.0	6.05	0.81
23XQ16	3	−5.17	0.02	−1.12	0.01	0.525	0.019	0.003	0.007	70.5	5.2	9.31	0.81
23XQ139	3	−3.43	0.01	−4.89	0.00	0.525	0.011	0.004	0.007	70.4	5.2	5.49	0.68
23XQ212	3	−4.56	0.01	−4.91	0.00	0.518	0.019	0.003	0.007	73.9	5.3	6.00	0.80
23XQ240	2	2.23	0.01	−0.56	0.00	0.547	0.007	0.004	0.008	60.1	5.1	8.23	0.86
23XQ257	3	−3.67	0.04	−1.76	0.01	0.541	0.031	0.004	0.007	62.7	4.9	7.45	0.82
23XQ261	3	−5.67	0.02	−2.05	0.00	0.576	0.007	0.003	0.007	47.4	4.3	4.58	0.82

Note: Rep. no. = replicate number; SD = standard deviation; SE = standard error; 95% conf. = 95% confidence interval; Temp.= temperature. Complete dataset can be found in Supplementary Dataset S1.

insufficient to elevate the temperatures of calcite and dolomite from ~ 25 °C to 47–77 °C (Fig. 7A, C). Assuming the measured $T_{(\Delta 638)}$ is a product of solid-state reordering, the maximum burial temperature should be higher than 170 °C for calcite and 200 °C for dolomite (Hemingway and Henkes, 2021; Fig. 7B, D).

The kinetic isotope effects of oxygen and clumped isotopes can be induced by microbial sulfate reduction (MSR) during the formation of dolomite, as MSR can mediate the pH and DIC concentrations of dolomitizing fluids (Lu and Swart, 2024; Li et al., 2024b). If this is the case, the kinetic isotope effect would produce a positive disequilibrium effect to clumped isotopes and a negative disequilibrium effect on $\delta^{18}O_c$ values (Guo, 2020). However, the correlation of Δ_{638} and δ^{18}Oc values in this study is ambiguous ($R^2 = 0.2$) (Fig. 8A) and inconsistent with the pattern of kinetic isotope effects.

Alternatively, water-rock interaction most likely accounts for the elevated clumped isotopic temperatures. The micrites contain higher concentrations of U (3.2–30.2 ppm) and Th (mostly 1.3–8.1 ppm, with four samples at 0.4–0.7 ppm) than typical lake water-precipitated carbonates, which usually have U and Th concentrations < 1 ppm (Elisha et al., 2021; Smedley and Kinniburgh, 2023). Such enrichment is indicative of interaction with diagenetic fluids (Li et al., 2024a, 2024c). In addition, Mn/Sr ratioss of > 1 in most samples are also indicative of diagenetic modification, as this ratio is typically < 1 in primary carbonates (Schobben et al., 2016; Table S1). Other geochemical proxies, including slightly positive Eu anomaly and MREE enrichment, together with dull to bright cathodoluminescence which suggests relatively high concentrations of Mn^{2+} (Pierson, 1981), also support fluid alteration.

Micritic to laminated texture and homogeneous distribution of Mg in the studied samples are consistent with dolomitization through cryptic (rather than macro-scale) recrystallization. The variable stoichiometry of the dolomites (42.4–49.2 mol% $MgCO_3$; lower than typical ones) and their low cation ordering values (0.26 to 0.47), indicate negligible effect of deep burial recrystallization and that the dolomites probably formed in the shallow burial stage (Manche and Kaczmarek, 2021). The lack of clear correlations between dolomite stoichiometry and cation ordering, and between these parameters and clumped-isotope temperature or carbonate δ^{18}O values (Fig. S2), further argues against temperature-driven deep burial recrystallization. Instead, the crystal-chemical maturity of the dolomites was likely controlled by multiple factors, including fluid composition and water-rock interaction.

The measured ~47–77 °C clumped-isotope temperatures of carbonates, most of which are >700 m above the basal tuff, are broadly compatible with the burial thermal regime inferred from apatite

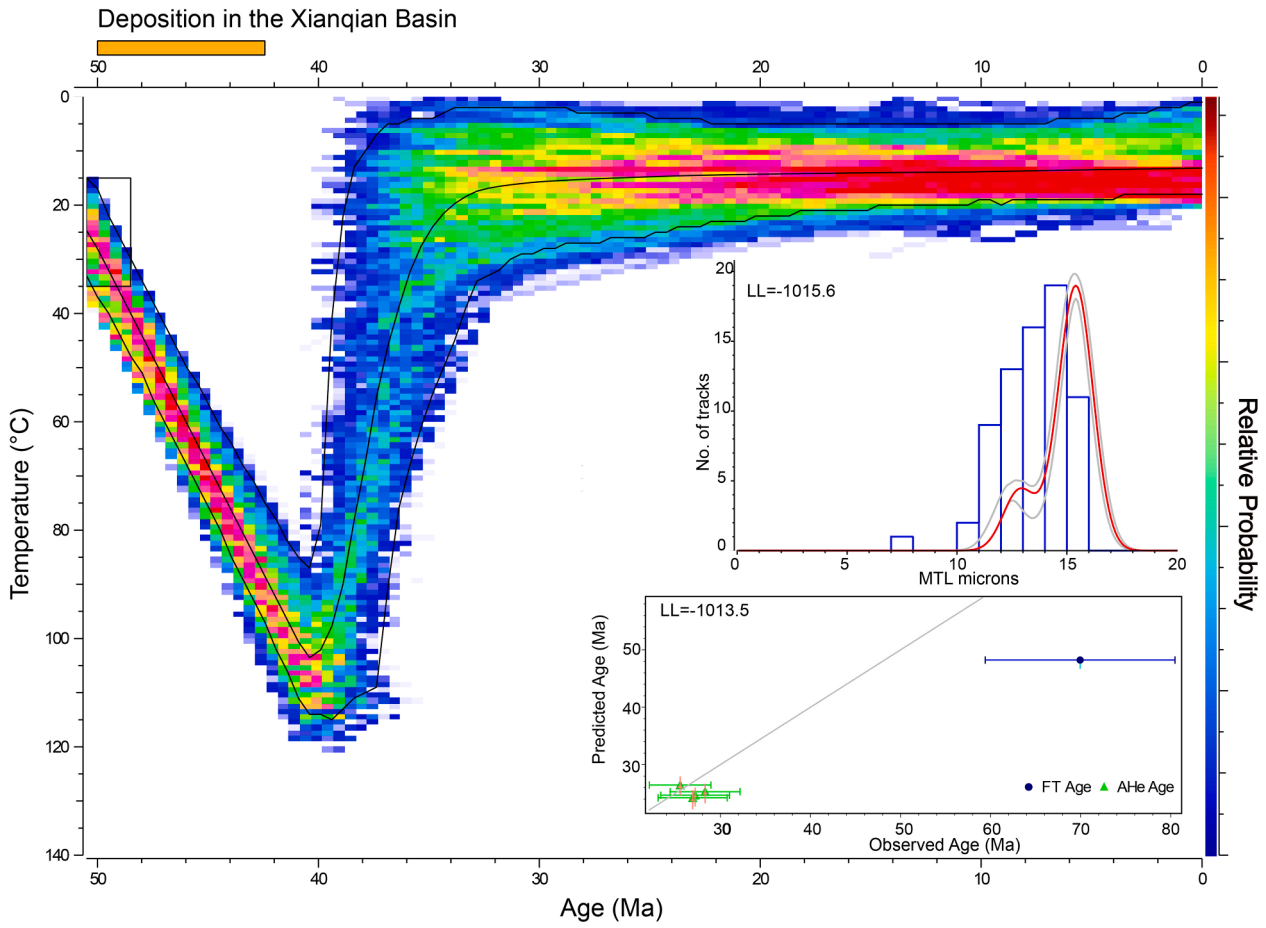


Fig. 6. Expected thermal history model (black line) with 95% confidence interval for the combined tuff (23GZ60) and sandstone (23GZ61) samples near the base of the Maerguo Formation. The constraints were justified in Table S6. The insets show predicted versus observed ages and fission track length, respectively. FT – fission-track; MTL – mean track length; LL – log-likelihood.

thermochronology. This suggests that the dolomitizing fluids probably have approached equilibrium with the burial geothermal gradient (Fig. 6). The upward decrease trend in clumped-isotope temperatures, although not strong (Fig. 5D), is broadly consistent with reduced thermal overprinting higher in the section, i.e., lower burial temperatures up section.

5.3. Oxygen isotopes of dolomitization fluid and implication for the paleoelevation studies

Micritic, massive, or laminated carbonates, such as those observed in the Xianqian and late Eocene Gerze basins, are commonly interpreted as textural indicators that a stratigraphic record may preserve primary $\delta^{18}O_c$ values (Fig. 4; Wei et al., 2016; Ingalls et al., 2020). The relatively positive $\delta^{18}O_c$ values (between −6 and −2‰), as observed in the Xianqian (Fig. 8B) and Gerze basins, could be interpreted to indicate either deposition in ocean water (Wei et al., 2016) or evaporated lake water (Ingalls et al., 2020). However, the induced dolomitization at burial temperatures (47–77 °C) in the Xianqian Basin (this study) and the Gerze Basin (Li et al., 2024a) indicates that the carbonates may not have retained their original $\delta^{18}O_c$ signature. Using the measured $T_{(\Delta 638)}$ temperatures and $\delta^{18}O_c$ values, accounting for the relative proportion of dolomite and calcite, the $\delta^{18}O_w$ values of the diagenetic fluid were estimated to range between +1‰ and +6‰ (Fig. 5E). These $\delta^{18}O_w$ values are more positive than those of modern evaporated lake water in south-central Tibet, which is between −3‰ and −9‰ (Li and Garzione, 2017).

Most samples plot, within error, near the –2‰ (VPDB) constant $\delta^{18}O_c$ contour in the diagram of clumped isotope temperatures $T_{(\Delta 638)}$ versus diagenetic fluid $\delta^{18}O_w$ values (Fig. 8C); such a distribution is usually interpreted to be diagnostic of rock-buffered recrystallization or solid-state reordering, which would keep the $\delta^{18}O_c$ values unchanged (e.g., Henkes et al., 2014; Huntington et al., 2015). However, the solid-state reordering is unlikely as the maximum burial temperature cannot cause bond reordering (Section 5.2). Alternatively, the alignment of $T_{(\Delta 638)}$ and $\delta^{18}O_w$ along $\delta^{18}O_c$ contour may haveresulted from various degrees of diagenetic fluid-mediated alteration due to covariance of fluid temperature and $\delta^{18}O_w$ values (Huntington and Lechler, 2015).

Intense evaporation, dissolution of evaporite minerals (e.g., gypsum and halite), and compactional expulsion may contribute to the ^{18}O-enriched basin brine as the diagenetic fluid. Alternatively, the diagenetic fluid may also originate from magmatic hydrothermal fluid with high $\delta^{18}O_w$ values (>+2‰; Bindeman, 2008), considering the proximal syn-depositional volcanism as suggested by the abundant tuff beds in the Xianqian Basin. Such diagenetic fluid has been revealed in other tuffaceous lacustrine basins, including Paleocene-Eocene Linzhou Basin in southern Tibet (Huang et al., 2022) and Late Paleozoic Junggar Basin in northwest China (e.g., Cao et al., 2020; Guo et al., 2023). Future work on fluid inclusions, mineralogy, geochemistry, or other independent indicators, can further constrain the nature of diagenetic fluids.

Given that the carbonates in the Xianqian Basin have undergone significant burial-related dolomitization, their $\delta^{18}O$ signals may have been distinct from those of the original lake water and thus should be used cautiously when used for paleoelevation reconstruction. This case is comparable to similar findings from the Gerze Basin (Li et al., 2024a) (Table 2). In addition, non-marine carbonates in the Nianbo and Pana

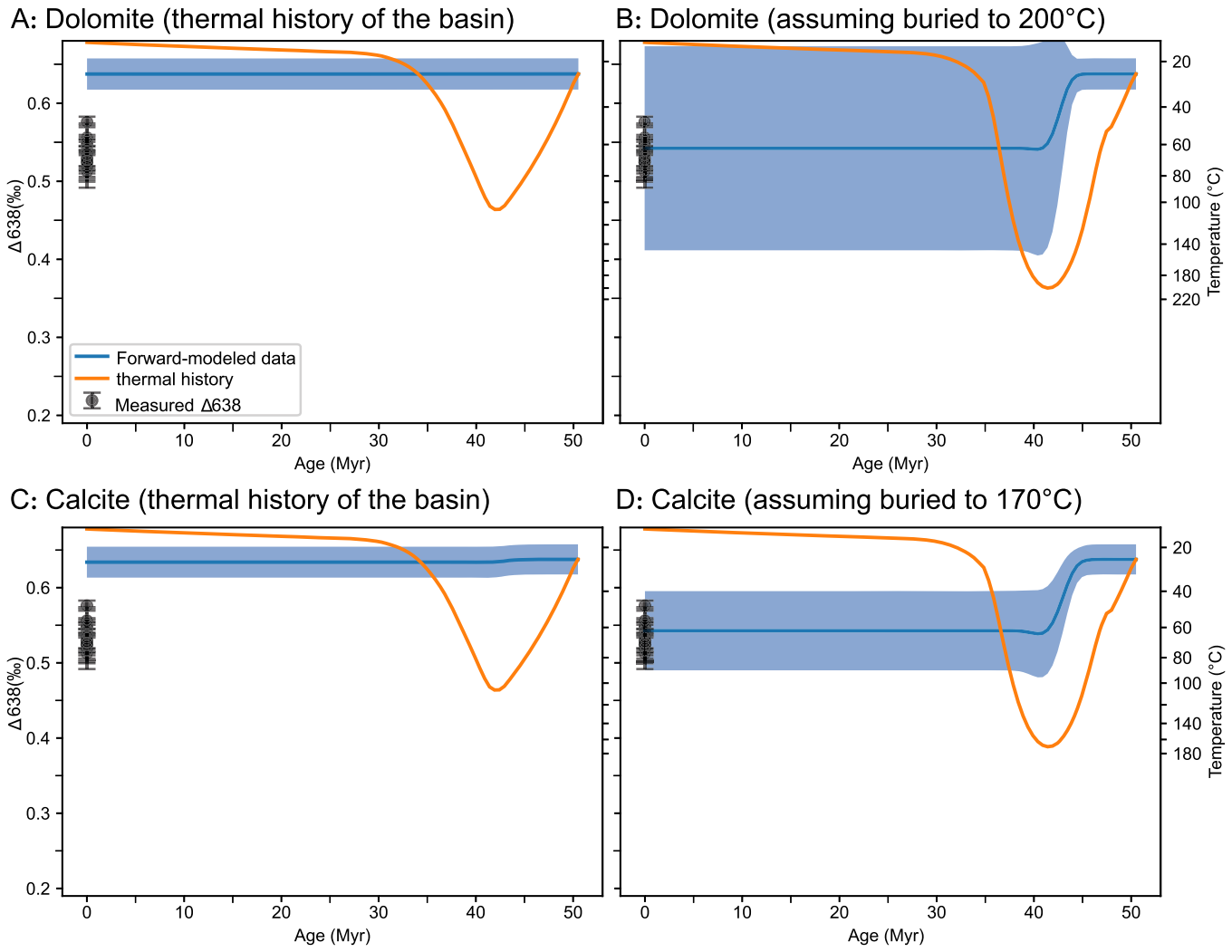


Fig. 7. Results of solid-state bond reordering modeling for carbonate minerals that initially formed at 25 °C. (A, C) For dolomite and calcite subjected to the thermal history reconstructed from AHe and AFT data, with a maximum burial temperature of ~106 °C, no significant clumped isotope bond reordering is recorded; (B, D) to induce solid-state bond reordering of clumped isotope to a temperature of ~60–70 °C, the maximum burial temperature of 200 °C and 170 °C is needed for dolomite and calcite, respectively.

formations of the Linzhou Basin, yielding clumped isotope temperatures of 6.3–105 °C, have been heated to 300 °C (54–50 Ma) and 130–170 °C (42–27 Ma), therefore they probably did not retain their original $\delta^{18}O_c$ values (Quade et al., 2020; Huang et al., 2022) (Table 2). In contrast, samples from other Paleogene basins in Tibet that experienced only solid-state bond reordering (e.g., Nima) may preserve original $\delta^{18}O_c$ values (DeCelles et al., 2007; Li et al., 2024c), while samples with clumped isotope temperatures below 35 °C (e.g., Lunpola, Nangqian, and Gongjue) might retain both depositional temperatures and $\delta^{18}O_c$ values (Li et al., 2019; Xiong et al., 2020, 2022; Li et al., 2024b); such samples are suitable for paleoelevation reconstruction (Table 2).

We therefore emphasize that lacustrine carbonates in tectonically active orogenic plateaus such as Tibet must be carefully evaluated for their diagenetic histories before being used in paleoaltimetry studies. In particular, because no thermochronology, clumped isotope, or elemental concentration data have yet been applied to assess carbonate diagenesis in the Heihuling Basin of the northern Qiangtang terrane (Table 2), the previously proposed conclusion that this region had reached near-present elevation by 51–28 Ma (Xu et al., 2013) warrants re-evaluation.

6. Conclusions

Our study applied apatite U-Th/He and fission-track thermochronology to reconstruct the thermal history of the Eocene alluvial to lacustrine Xianqian Basin in central Tibet. The Xianqian Basin records deposition from 50 to 43 Ma, burial to a depth sufficient to reset AHe ages and partially anneal AFT ages, and finally exhumation from >2 km depth to near-surface conditions by ~34 Ma. Eocene sediment burial caused concurrent fluid-mediated dolomitization of lacustrine carbonates, resulting in carbonate clumped-isotope temperatures of 47–77 °C. Calculated $\delta^{18}O$ values of the diagenetic fluids, ranging from +1‰ to +6‰, cannot be used to constrain paleoelevation. Our study highlights the important role of basin thermal history reconstruction for determining carbonate diagenesis history, and the need to rigorously assess potential diagenetic alteration prior to using lacustrine carbonate $\delta^{18}O_c$ to reconstruct paleoelevation.

CRediT authorship contribution statement

Anlin Ma: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Conceptualization. **Lin Li:** Writing – review & editing, Methodology, Conceptualization. **Guangwei Li:** Writing – review & editing, Validation, Resources, Methodology. **John**

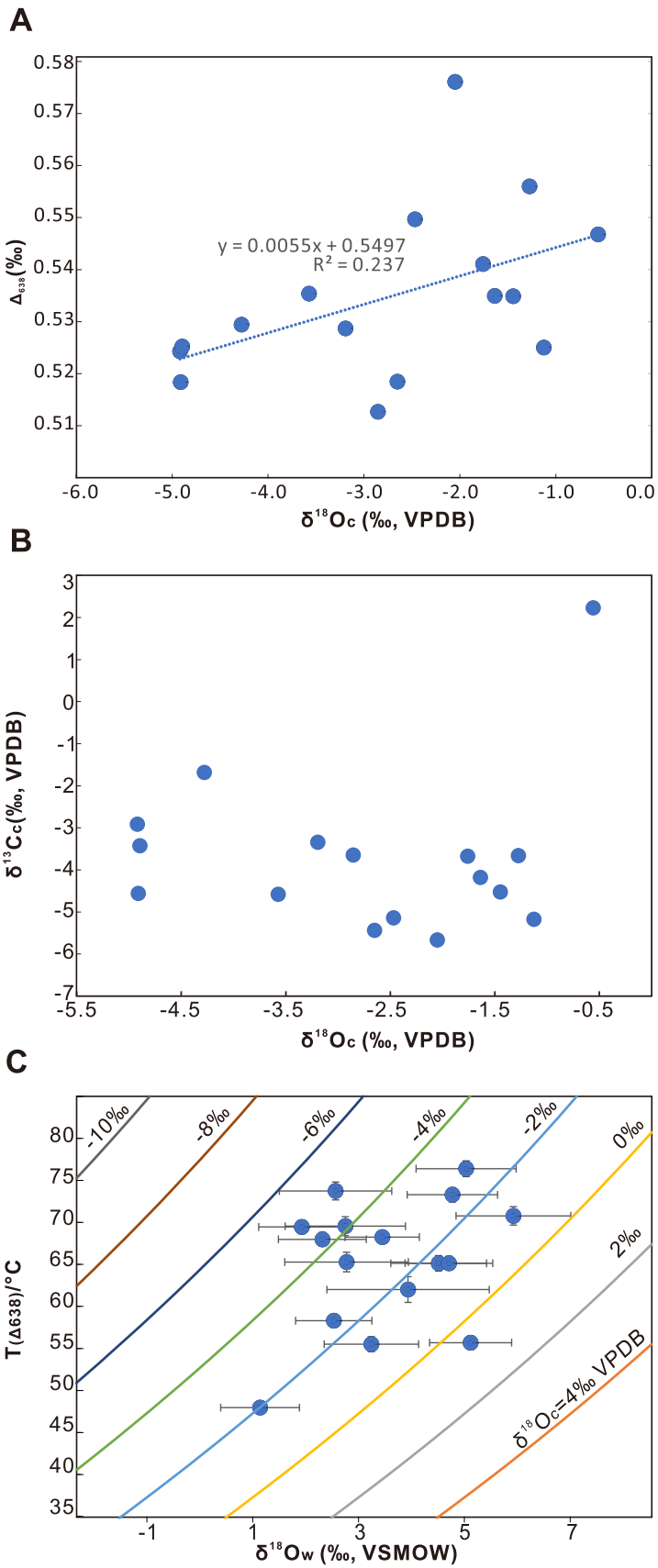


Fig. 8. Interpretation of carbonate $\delta^{18}O_c$ data and diagenetic fluid origins. (A) Cross-plots of Δ_{638} and $\delta^{18}O_c$ values. (B) Cross-plots of $\delta^{13}C_c$ and $\delta^{18}O_c$ values. (C) Clumped isotope temperatures $T_{(\Delta 638)}$ versus calculated $\delta^{18}O_w$ values of the diagenetic fluid, which were calculated using the mineral-water oxygen isotope thermometry equation of Horita (2014) and Kim and O'Neil (1997) for dolomite and calcite, respectively, as in Dataset S1. Contours represent constant $\delta^{18}O_c$ values based on the dolomite-water oxygen isotope thermometry (Horita, 2014).

Table 2
Summary of carbonate diagenesis in Paleogene terrestrial basins in Tibet.

Basin*	Dolomitization	recrystallization	solid-state bond reordering	Thermochronology	Maximum burial	Clumped isotope temperature	Reliability for paleoelevation
Linzhou (~60–50 Ma)	⋄	⋄	⋄	ZHe, AHe, and $^{40}Ar/^{39}Ar$	300 °C at 54–50 Ma and 130–170 °C at 42–27 Ma	6.3–105 °C	$\delta^{18}O$ probably not primary
Gerze (~37 Ma)	■	□	□	⋄	<~2 km; 40–50 °C	30–50 °C	$\delta^{18}O$ not primary
Nima (~25 Ma)	■	■	■	⋄	>~2.8 km	30–130 °C	$\delta^{18}O$ preserved in samples only with bond reordering
Lunpola (~38 Ma)	□	■	□	⋄	~2.8 km	12–64 °C	Primary temperature & $\delta^{18}O$ in some samples
Xianqian (49–43 Ma)	■	□	□	AHe and AFT	2 km; 106 °C at 42 Ma	47–77 °C	$\delta^{18}O$ not primary
Heihuling (51–28 Ma)	⋄	⋄	⋄	⋄	⋄	⋄	Further evaluation needed
Gongjue (50–40 Ma)	□	■	□	⋄	2–3 km	8–86 °C	Primary temperature & $\delta^{18}O$ in some samples
Nangqian (Late Cretaceous to 37 Ma)	⋄	⋄	⋄	⋄	2–3 km	21–77 °C	Primary temperature & $\delta^{18}O$ in some samples

Note: * Linzhou Basin: Ingalls et al., 2018; Quade et al., 2020; Huang et al., 2022; Gerze Basin: Li et al., 2024a; Nima Basin: DeCelles et al., 2007; Huntington et al., 2015; Li et al., 2024c; Lunpola Basin: Ingalls et al., 2020; Xiong et al., 2022; Li et al., 2024b; Xianqian Basin: this study; Heihuling Basin: Xu et al., 2013; Gongjue Basin: Li et al., 2019; Nangqian Basin: Xiong et al., 2020. □: absent; ■: present; ⋄: Not determined.

J.Y. He: Writing – review & editing, Validation, Conceptualization. **Paul Kapp:** Writing – review & editing, Validation, Funding acquisition. **Chaojin Lu:** Writing – review & editing, Visualization, Software, Conceptualization. **Hong Zhang:** Writing – review & editing, Resources, Methodology. **Cannan Lin:** Methodology, Writing – review & editing. **Jordan W. Wang:** Writing – review & editing. **Bihai Zheng:** Writing – review & editing. **Keran Li:** Visualization, Writing – review & editing. **Xiumian Hu:** Writing – review & editing, Validation, Project administration, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Anlin Ma reports financial support was provided by National Natural Science Foundation of China. J.J.Y. He, P. Kapp, and J.W. Wang report financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2026.120328.

Data availability

Data will be made available on request.

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